Math 105L In Class 3 Practice Solutions

Contents

Derivatives of Logs 2

Using First and Second Derivatives 4

Global Max's and Min's 9

Optimization 13

Linear Approximations 16

L'Hopital's Rule 18

Related Rates 22

Derivatives of logs

1. Differentiate $y = \ln(x^2 + 1)$.

   We use the chain rule. Let $u = x^2 + 1$ so $y = \ln u$.

   $$\frac{dy}{dx} = \frac{1}{u}\frac{du}{dx} = \frac{1}{x^2+1} \cdot 2x = \frac{2x}{x^2+1}.$$

2. Differentiate $y = \log(3x)$.

   Use the derivative formula for base-10 logarithm: $\frac{d}{dx}\log v = \frac{1}{v \ln 10}\frac{dv}{dx}$. Here $v = 3x$ and $\frac{dv}{dx} = 3$.

   $$\frac{dy}{dx} = \frac{1}{3x \ln 10} \cdot 3 = \frac{1}{x \ln 10}.$$

3. Differentiate $y = \ln\left(1 + e^{2x}\right)$.

   Chain rule with $u = 1 + e^{2x}$:

   $$\frac{dy}{dx} = \frac{1}{1+e^{2x}} \cdot \frac{d}{dx}(e^{2x}) = \frac{1}{1+e^{2x}} \cdot 2e^{2x} = \frac{2e^{2x}}{1+e^{2x}}.$$

4. Differentiate $y = (\ln x)^2$ for $x > 0$.

   Use the chain rule: let $u = \ln x$, then $y = u^2$.

   $$\frac{dy}{dx} = 2u\frac{du}{dx} = 2\ln x \cdot \frac{1}{x} = \frac{2\ln x}{x}.$$

5. Differentiate $y = \ln\left(\frac{x^2}{\sqrt{1+x}}\right)$ for $x > 0$.

   Simplify using log laws first:

   $$y = \ln(x^2) - \ln\left((1+x)^{1/2}\right) = 2\ln x - \tfrac{1}{2}\ln(1+x).$$

   Now differentiate termwise:

   $$\frac{dy}{dx} = 2 \cdot \frac{1}{x} - \tfrac{1}{2} \cdot \frac{1}{1+x} = \frac{2}{x} - \frac{1}{2(1+x)}.$$

   If desired, combine over a common denominator:

   $$\frac{dy}{dx} = \frac{4(1+x) - x}{2x(1+x)} = \frac{4+3x}{2x(1+x)}.$$

6. Differentiate $y = \ln\left(\sqrt{x^2+4}\right)$.

   Note $\sqrt{x^2+4} = (x^2+4)^{1/2}$. Then $y = \frac{1}{2}\ln(x^2+4)$. Differentiate:

   $$\frac{dy}{dx} = \tfrac{1}{2} \cdot \frac{1}{x^2+4} \cdot 2x = \frac{x}{x^2+4}.$$

7. Differentiate $y = \ln(x) \cdot \ln(2x)$ for $x > 0$.

Product rule: if $u = \ln x$ and $v = \ln(2x)$ then $u' = \frac{1}{x}$ and $v' = \frac{1}{x}$.

$$\frac{dy}{dx} = u'v + uv' = \frac{1}{x}\ln(2x) + \ln x \cdot \frac{1}{x} = \frac{\ln(2x) + \ln x}{x}.$$

Use log property $\ln(2x) + \ln x = \ln(2x^2)$ if you want a single log:

$$\frac{dy}{dx} = \frac{\ln(2x^2)}{x}.$$

8. Differentiate $y = \ln\left(x + \sqrt{1+x^2}\right)$.

Let $u = x + \sqrt{1+x^2}$. Differentiate using chain rule. First compute $\frac{du}{dx} = 1 + \frac{1}{2}(1+x^2)^{-1/2} \cdot 2x = 1 + \frac{x}{\sqrt{1+x^2}}$. Then

$$\frac{dy}{dx} = \frac{1}{u}\frac{du}{dx} = \frac{1 + \frac{x}{\sqrt{1+x^2}}}{x + \sqrt{1+x^2}}.$$

This simplifies to the neat form

$$\frac{dy}{dx} = \frac{1}{\sqrt{1+x^2}}.$$

9. Differentiate $y = \log(x^3 + 1)$.

Use the derivative formula for base-10 logarithm: $\frac{d}{dx}\log v = \frac{1}{v \ln 10}\frac{dv}{dx}$. Here $v = x^3 + 1$ and $\frac{dv}{dx} = 3x^2$.

$$\frac{dy}{dx} = \frac{1}{(x^3+1)\ln 10} \cdot 3x^2 = \frac{3x^2}{(x^3+1)\ln 10}.$$

10. Differentiate $y = \ln\big(f(x)\big)$ where $f(x) = x^2 e^{3x}(x+1)$ (assume x where $f(x) > 0$).

Use log properties to simplify:

$$y = \ln(x^2) + \ln(e^{3x}) + \ln(x+1) = 2\ln x + 3x + \ln(x+1).$$

Differentiate termwise:

$$\frac{dy}{dx} = 2 \cdot \frac{1}{x} + 3 + \frac{1}{x+1} = \frac{2}{x} + 3 + \frac{1}{x+1}.$$

Alternatively, one could use the chain rule directly: $\frac{dy}{dx} = \frac{f'(x)}{f(x)}$, which yields the same result after simplifying.

Using First and Second Derivatives

1. For each of the following, find and classify all critical points. Find all inflection points too.

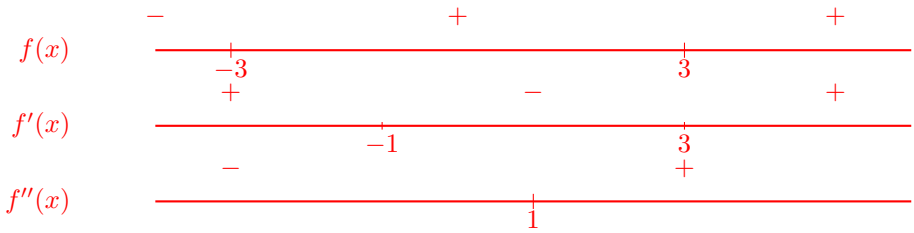
   (a) $f(x) = x^3 - 3x^2 - 9x + 27 = (x-3)^2(x+3)$

   Original form: $f(x) = x^3 - 3x^2 - 9x + 27$
   Factored form: $f(x) = (x-3)^2(x+3)$
   Roots: $x = -3, 3$

   Critical points: $f'(x) = 3x^2 - 6x - 9 = 3(x-3)(x+1) \implies x = -1, 3$

   Potential inflection points: $f''(x) = 6x - 6 = 6(x-1) \implies x = 1$
   Since $f''(x)$ changes sign at $x = 1$, we confirm an inflection point at $x = 1$.

   Number lines:

   Max/Min: $x = -1$ maximum, $x = 3$ minimum
   Inflection point: $x = 1$, $f(1) = 16$

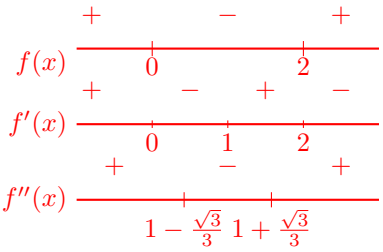
   (b) $f(x) = x^4 - 4x^3 + 4x^2 = x^2(x-2)^2$

   Original form: $f(x) = x^4 - 4x^3 + 4x^2$
   Factored form: $f(x) = x^2(x-2)^2$
   Roots: $x = 0, 2$

   Critical points: $f'(x) = 4x^3 - 12x^2 + 8x = 4x(x-2)(x-1) \implies x = 0, 1, 2$

   Potential inflection points: $f''(x) = 12x^2 - 24x + 8 = 4(3x^2 - 6x + 2) = 0 \implies x = 1 \pm \frac{\sqrt{3}}{3}$
   Since $f''(x)$ changes sign at $x = 1 \pm \frac{\sqrt{3}}{3}$, these are confirmed inflection points.

   Number lines:

   Max/Min: $x = 0$ minimum, $x = 2$ minimum, $x = 1$ maximum
   Inflection points: $x = 1 \pm \frac{\sqrt{3}}{3}$

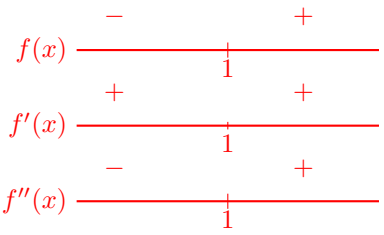
   (c) $f(x) = x^3 - 3x^2 + 3x - 1 = (x-1)^3$

   Original form: $f(x) = x^3 - 3x^2 + 3x - 1$
   Factored form: $f(x) = (x-1)^3$
   Roots: $x = 1$

   Critical points: $f'(x) = 3(x-1)^2 = 0 \implies x = 1$

   Potential inflection points: $f''(x) = 6(x-1) = 0 \implies x = 1$
   $f''(x)$ changes sign at $x = 1$, so this is an **inflection point**.

Number lines:



Max/Min: none (horizontal tangent at $x = 1$)

Inflection points: $x = 1$

(d) $f(x) = x^4 - 2x^2 = x^2(x - \sqrt{2})(x + \sqrt{2})$

Original form: $f(x) = x^4 - 2x^2$

Factored form: $f(x) = x^2(x - \sqrt{2})(x + \sqrt{2})$

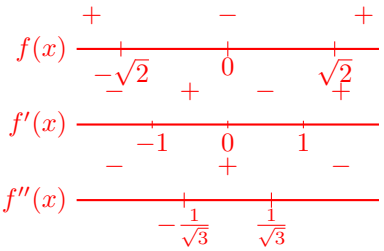
Roots: $x = -\sqrt{2}, 0, \sqrt{2}$

Critical points: $f'(x) = 4x^3 - 4x = 4x(x-1)(x+1) \implies x = -1, 0, 1$

Potential inflection points: $f''(x) = 12x^2 - 4 = 0 \implies x = \pm\frac{1}{\sqrt{3}}$

$f''(x)$ changes sign at these points, so both are confirmed inflection points.

Number lines:



Max/Min: $x = -1$ maximum, $x = 0$ minimum, $x = 1$ maximum

Inflection points: $x = \pm\frac{1}{\sqrt{3}}$

(e) $f(x) = x^3 - 3x = x(x - \sqrt{3})(x + \sqrt{3})$

Original form: $f(x) = x^3 - 3x$

Factored form: $f(x) = x(x - \sqrt{3})(x + \sqrt{3})$

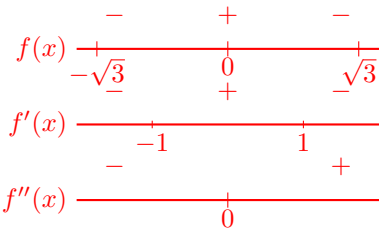
Roots: $x = -\sqrt{3}, 0, \sqrt{3}$

Critical points: $f'(x) = 3x^2 - 3 = 3(x-1)(x+1) \implies x = -1, 1$

Potential inflection points: $f''(x) = 6x = 0 \implies x = 0$

Since $f''(x)$ changes sign at $x = 0$, we confirm an inflection point at $x = 0$.

Number lines:



Max/Min: $x = -1$ maximum, $x = 1$ minimum

Inflection point: $x = 0$

(f) $f(x) = xe^x - e^x = e^x(x - 1)$

Original form: $f(x) = xe^x - e^x$

Factored form: $f(x) = e^x(x - 1)$

Roots: $f(x) = 0 \implies e^x(x-1) = 0 \implies x = 1$

(Note: $e^x \neq 0$ for any real x)

Critical points:

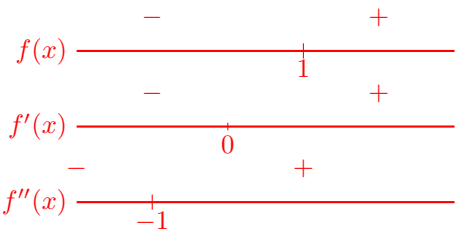
$$f'(x) = e^x + xe^x - e^x = xe^x \implies xe^x = 0 \implies x = 0$$

Potential inflection points:

$$f''(x) = e^x + xe^x = e^x(x+1) \implies x = -1$$

Since $f''(x)$ changes sign at $x = -1$, we confirm an inflection point at $x = -1$.

Number lines:



Max/Min: $x = 0$ minimum

Inflection point: $x = -1$, $f(-1) = -2/e$

2. Give examples of functions satisfying each of the following sets of conditions:

   (a) $f'(0) = 0$, but $f'(x)$ does not have a max or min at $x = 0$

   We need a critical point that isn't a max or a min.

   Eg $f(x) = x^3$.

   (b) $f'(0) = 0$, $f''(0) = 0$, and $f'(x)$ has a max at $x = 0$.

   $f(x) = -x^4$

   (c) $f''(x) = 0$, but $f(x)$ does not have an IP at $x = 0$.

   $f(x) = x^4$

   (d) $f(x)$ has a min at $x = 0$, but $f'(0) \neq 0$.

   We need a function where $f'(x)$ is undefined at $x = 0$, but has a min there.

   Eg $f(x) = |x|$

   (e) $f''(0) > 0$, but $f(x)$ does not have a min at $x = 0$.

   Any function that is concave up at $x = 0$, but doesn't have a critical point there. For example, $f(x) = (x-1)^2$.

   (f) $f'(0)$ is defined at $x = 0$, $f''(0)$ is not defined, and $f(x)$ has an IP at $x = 0$

   (This last one is hard – skip it if you want. It will not be on the test.)

   $$f(x) = \begin{cases} x^2 & x \geq 0 \\ -x^2 & x < 0 \end{cases}$$

   The derivative here would be

   $$f'(x) = \begin{cases} 2x & x > 0 \\ -2x & x < 0 \end{cases}$$

   or $f'(x) = 2|x|$ away from $x = 0$.

   We can pick $f'(0) = 0$ to make this continuous and therefore make the original function differentiable.

   $f'(x) = 2|x|$ is not differentiable at $x = 0$, so $f''(0)$ DNE, but

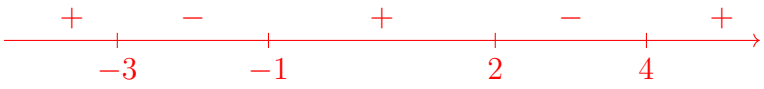
   $$f''(x) = \begin{cases} 2 & x > 0 \\ -2 & x < 0 \end{cases}$$

   which changes sign, so there's an IP at $x = 0$.

3. In each part below, number lines are presented for the first and second derivatives of a function. Find and classify all critical points. Find all inflection points.

   Also: precisely one of these has an error where the classification is logically inconsistent. Find it and explain.

(a) $f'(x)$



$f''(x)$

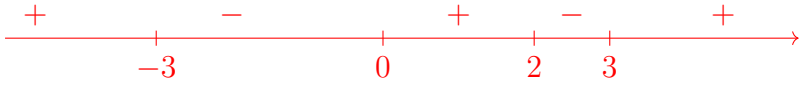


CPs: $x = -3$ (local max), $x = -1$ (local min), $x = 2$ (local max), $x = 4$ (local min). IPs: $x = -2$, 1, 3.

(b) $f'(x)$



$f''(x)$



CPs: $x = -4$ (local min), $x = -2$ (local max), $x = 1$ (local min), $x = 3$ (neither). IPs: $x = -3$, 0, 2.

(c) $f'(x)$



$f''(x)$



CPs: $x = -3$ (local min), $x = 0$ (local max), $x = 2$ (local min). IPs: $x = -2$, 1, 3.

(d) $f'(x)$



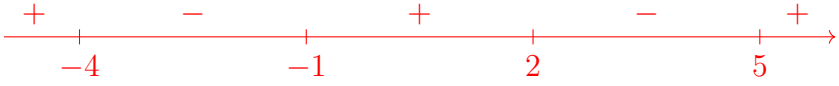
$f''(x)$

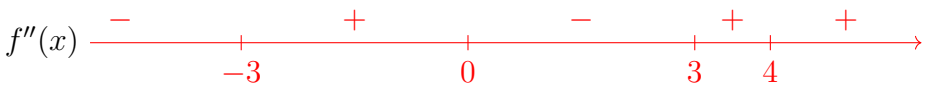


CPs: $x = -2$ (local max), $x = 1$ (local min), $x = 4$ (neither). IPs: $x = -3$, 0, 2, 3.

This is logically inconsistent, as the max at $x = -2$ has a $f'(-2) > 0$, and the min at $x = 1$ has $f''(1) < 0$.

(e) $f'(x)$





Critical points: $x = -4$ (local max), $x = -1$ (local min), $x = 2$ (local max), $x = 5$ (local min).
Inflection points: $x = -3$, 0, 3.
Note: $x = 4$ is a zero of f'' but not an inflection point (no sign change at $x = 4$), which allows $f''(5) > 0$ so $x = 5$ can be a local minimum.

Global Max's and Min's

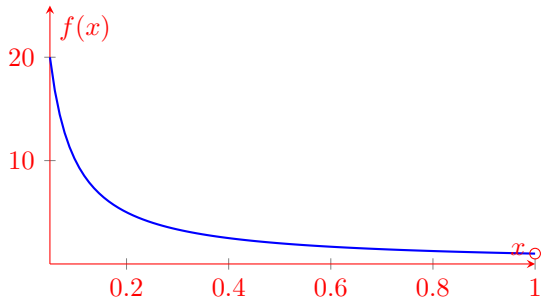
For each of the following functions, find its global maximum and/or global minimum on the given domain, if they exist. Pay careful attention to cases where the domain is not closed or not finite!

1. $f(x) = \dfrac{1}{x}$ for $0 < x < 1$.

   $f'(x) = -\dfrac{1}{x^2} < 0$, so f is strictly decreasing.

   As $\lim_{x \to 0^+} f(x) = +\infty$ and $\lim_{x \to 1^-} f(x) = 1$, the function decreases toward 1 but never reaches it.

   Hence there is **no global maximum** and **no global minimum**.



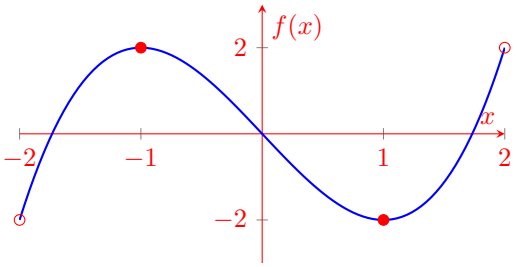
2. $f(x) = x^3 - 3x$ for $-2 < x < 2$.

   $f'(x) = 3x^2 - 3 = 3(x-1)(x+1)$, so critical points are $x = -1, 1$.

   $f(-1) = 2$, $f(1) = -2$.

   $\lim_{x \to -2^+} f(x) = -2$ and $\lim_{x \to 2^-} f(x) = 2$.

   Hence **global max** is 2 at $x = -1$, **global min** is -2 at $x = 1$.



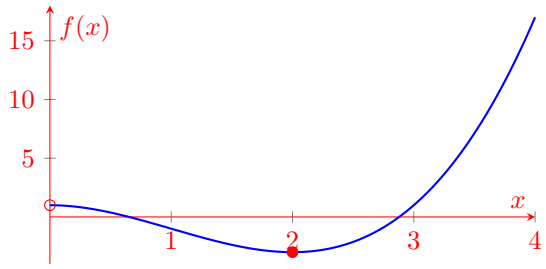
3. $f(x) = x^3 - 3x^2 + 1$ for $0 < x < 4$.

   $f'(x) = 3x(x-2)$, critical point at $x = 2$.

   $f(2) = -3$.

   $\lim_{x \to 0^+} f(x) = 1$, $\lim_{x \to 4^-} f(x) = 17$.

   Global minimum is -3 at $x = 2$, **no global maximum** since $x = 4$ is not included.

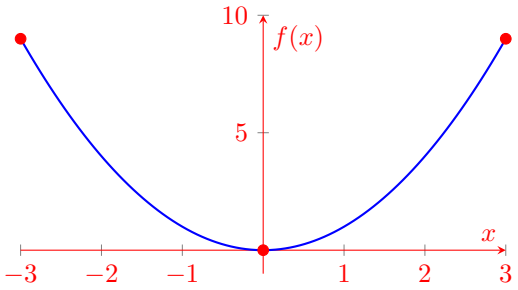


4. $f(x) = x^2$ for $-3 \leq x \leq 3$.

   $f'(x) = 2x = 0$ at $x = 0$.

   $f(0) = 0$, $f(-3) = f(3) = 9$.

   Global min 0 at $x = 0$, **global max** 9 at $x = -3$ and $x = 3$.

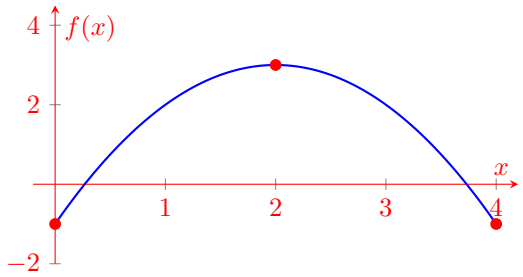


5. $f(x) = -x^2 + 4x - 1$ for $0 \leq x \leq 4$.

   $f'(x) = -2x + 4 = 0$ gives $x = 2$.

   $f(0) = -1$, $f(2) = 3$, $f(4) = -1$.

   Global max 3 at $x = 2$, **global min** -1 at $x = 0$ and $x = 4$.

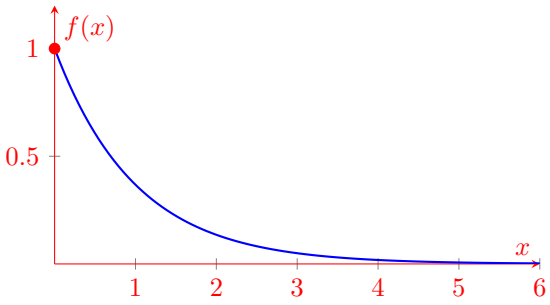


6. $f(x) = e^{-x}$ for $0 \leq x < \infty$.

   $f'(x) = -e^{-x} < 0$, so strictly decreasing.

   $f(0) = 1$, $\lim_{x \to \infty} f(x) = 0$.

   Global max 1 at $x = 0$, no global minimum (approaches 0).

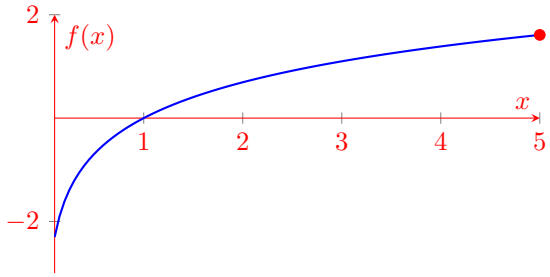


7. $f(x) = \ln(x)$ for $0 < x \leq 5$.

   $f'(x) = 1/x > 0$, so increasing.

   $\lim_{x \to 0^+} f(x) = -\infty$, $f(5) = \ln 5$.

   Global max $\ln 5$ at $x = 5$, no global minimum.



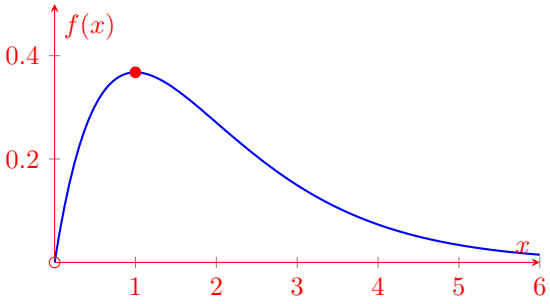
8. $f(x) = xe^{-x}$ for $0 < x < \infty$.

   $f'(x) = e^{-x}(1 - x)$, critical point at $x = 1$.

   $f(1) = e^{-1}$.

   $\lim_{x \to 0^+} f(x) = 0$, $\lim_{x \to \infty} f(x) = 0$.

   Global max e^{-1} at $x = 1$, no global minimum (approaches 0).



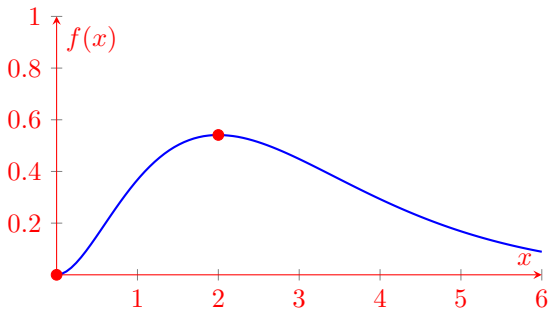
9. $f(x) = x^2e^{-x}$ for $0 \leq x < \infty$.

   $f'(x) = e^{-x}(2x - x^2) = e^{-x}x(2 - x)$, critical points at $x = 0, 2$.

   $f(0) = 0$, $f(2) = 4e^{-2}$.

   $\lim_{x \to \infty} f(x) = 0$.

   Global max $4e^{-2}$ at $x = 2$, **global min** 0 at $x = 0$.



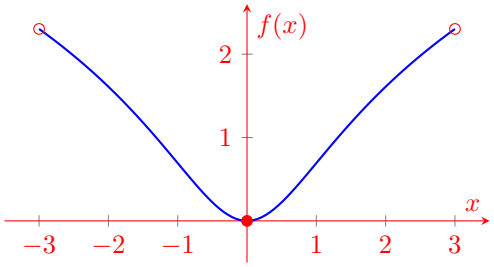
10. $f(x) = \ln(x^2 + 1)$ for $-3 < x < 3$.

$f'(x) = 2x/(x^2 + 1)$, critical point at $x = 0$.

$f(0) = 0$.

$\lim_{x \to -3^+} f(x) = \ln 10 \approx 2.3026$, $\lim_{x \to 3^-} f(x) = \ln 10 \approx 2.3026$.

Global min 0 at $x = 0$, no global maximum.



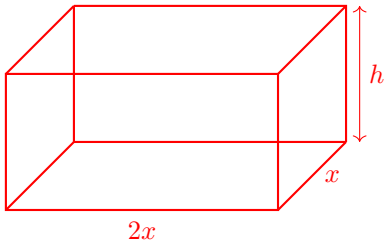
Optimization

Note that in these solutions, I used the second derivative test. We did more first derivative style tests in class. Those are also fine!

1. **Open-Top Wooden Storage Chest**

   A carpenter is building an *open-top storage chest* with a *rectangular base* whose *length is twice its width*. The chest must have a volume of 120 cubic inches. The wood for the base costs 5¢ per square inch, and the wood for the sides costs 2¢ per square inch.

   Find: The dimensions of the chest that minimize the total cost of materials.



Let the width be x and the length be $2x$. Then volume: $V = (2x)(x)(h) = 120$, so $h = \dfrac{60}{x^2}$.

Cost: The base costs $5¢/\text{in}^2$, area $2x^2$. Four sides: two of area xh, two of area $2xh$, total side area $6xh$. At $2¢/\text{in}^2$, side cost $= 12xh¢$.

So total cost

$$C = 5(2x^2) + 2(6xh) = 10x^2 + 12xh.$$

Substitute $h = 60/x^2$:

$$C(x) = 10x^2 + 12x\left(\frac{60}{x^2}\right) = 10x^2 + \frac{720}{x}.$$

Differentiate:

$$C'(x) = 20x - \frac{720}{x^2}.$$

Set $C'(x) = 0$:

$$20x^3 = 720 \implies x^3 = 36 \implies x = \sqrt[3]{36} \approx 3.30.$$

Check: $C''(x) = 20 + \frac{1440}{x^3} > 0$, confirming a minimum.

Then

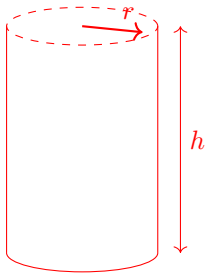
$$h = \frac{60}{x^2} = \frac{60}{(\sqrt[3]{36})^2} \approx 5.51.$$

Dimensions: Width ≈ 3.30 in, Length ≈ 6.60 in, Height ≈ 5.51 in.

2. **Open-Top Cylindrical Can**

   A manufacturer wants to design an open-top cylindrical can that will hold 500 cm^3 of liquid. The material for the sides costs 2¢ per cm^2, and the material for the base costs 5¢ per cm^2.

   Find: The radius and height of the can that minimize the total cost.



Volume: $V = \pi r^2 h = 500 \implies h = \frac{500}{\pi r^2}$.

Cost: Base area: πr^2 at $5¢/\text{cm}^2 \to 5\pi r^2$. Side area: $2\pi rh$ at $2¢/\text{cm}^2 \to 4\pi rh$. Total:

$$C = 5\pi r^2 + 4\pi rh.$$

Substitute $h = \frac{500}{\pi r^2}$:

$$C(r) = 5\pi r^2 + 4\pi r\left(\frac{500}{\pi r^2}\right) = 5\pi r^2 + \frac{2000}{r}.$$

Differentiate:

$$C'(r) = 10\pi r - \frac{2000}{r^2}.$$

Set $C'(r) = 0$:

$$10\pi r^3 = 2000 \implies r^3 = \frac{200}{\pi}.$$

Thus

$$r = \sqrt[3]{\frac{200}{\pi}} \approx 3.993 \text{ cm},$$

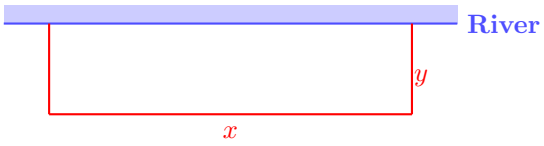
Minimum confirmed since $C''(r) = 10\pi + \frac{4000}{r^3} > 0$.

$$h = \frac{500}{\pi r^2} \approx 9.98 \text{ cm}.$$

3. **Fencing Along a River**

A rectangular field is to be fenced along a straight river. The side along the river requires no fence, and you have 100 m of fencing for the remaining three sides.

Find: The dimensions that maximize the enclosed area.



Let the length parallel to the river be x and the perpendicular side be y. You need fencing on the two sides of length y and the opposite side of length x:

$$2y + x = 100.$$

Area: $A = xy = x\left(\frac{100-x}{2}\right) = 50x - \frac{x^2}{2}$.

Differentiate:

$$A'(x) = 50 - x.$$

Set $A'(x) = 0 \Rightarrow x = 50$.

Maximum area: $A = 50 \cdot 25 = 1250 \text{ m}^2$. Second derivative $A''(x) = -1 < 0$, confirming a maximum.

Then $y = \frac{100-50}{2} = 25$.

Dimensions: 50 m along the river, 25 m deep.

4. **Two Numbers with Fixed Sum**

   Find two positive numbers whose sum is 12 and whose product is as large as possible.

   Let the numbers be x and $y = 12 - x$. Product $P = x(12 - x) = 12x - x^2$. Differentiate:

   $$P'(x) = 12 - 2x = 0 \Rightarrow x = 6.$$

   The function is concave down ($P'' = -2 < 0$), confirming a maximum.

   So $y = 6$.

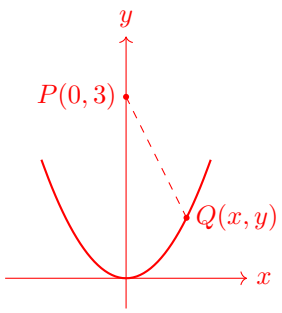
   Maximum product: $P = 36$.

   Conclusion: The two numbers are 6 and 6.

5. **Closest Point on a Parabola**

   Find the point $Q(x, y)$ on the parabola $y = x^2$ that is closest to the point $P = (0, 3)$.

   (Hint: The coordinates of a point on a parabola are (x, x^2). Find a formula for the square of the distance between P and this point, then minimize it.)



   Distance squared:

   $$D^2 = (x - 0)^2 + (x^2 - 3)^2 = x^4 - 5x^2 + 9.$$

   Differentiate:

   $$\frac{dD^2}{dx} = 4x^3 - 10x = 2x(2x^2 - 5).$$

   Set to 0:

   $$x = 0 \quad \text{or} \quad x^2 = \frac{5}{2}.$$

   Compute values:

   - $x = 0$: $D^2 = 9$.
   - $x = \sqrt{5/2} \approx 1.58$: $D^2 \approx 2.75$. (Note: negative root gives the same point, but on the other side of the parabola)

   Check min:

   $$\frac{d^2D^2}{dx^2} = 8x^2 - 10$$

   This is negative at $x = 0$ and positive at $x = \sqrt{5/2}$, so the latter is a min.

   Thus minimum at $x = \pm\sqrt{5/2}$. So

   $$Q\left(\pm\sqrt{\tfrac{5}{2}}, \tfrac{5}{2}\right)$$

   are the closest points, distance ≈ 1.66.

Linear Approximations

1. Approximate the value of $\sqrt[3]{27.1}$ using a linear approximation near $x = 27$.
Without computing the exact value of $\sqrt[3]{27.1}$, is your estimate an overestimate or underestimate? Explain your answer.

The function is $f(x) = x^{1/3}$, and its first derivative is $f'(x) = \frac{1}{3}x^{-2/3}$. Near $x = 27$, we have:

$$f(27) = 3, \quad f'(27) = \frac{1}{3}(27)^{-2/3} = \frac{1}{27}.$$

Using the linear approximation:

$$\sqrt[3]{27.1} \approx 3 + \frac{1}{27}(27.1 - 27) = 3 + \frac{1}{27}(0.1) \approx 3.0037.$$

The second derivative is $f''(x) = -\frac{2}{9}x^{-5/3}$, which is **negative** for $x > 0$. This means the function is concave down, indicating that the linear approximation will be an **overestimate**.

2. Use linear approximation to estimate the value of $\sqrt{49.2}$ by approximating the function $f(x) = \sqrt{x}$ near $x = 49$.
Without computing the exact value of $\sqrt{49.2}$, is your estimate an overestimate or underestimate? Explain your answer.

The function is $f(x) = \sqrt{x}$, and its first derivative is $f'(x) = \frac{1}{2\sqrt{x}}$. Near $x = 49$, we have:

$$f(49) = 7, \quad f'(49) = \frac{1}{2 \cdot 7} = \frac{1}{14}.$$

Using the linear approximation:

$$\sqrt{49.2} \approx 7 + \frac{1}{14}(49.2 - 49) = 7 + \frac{1}{14}(0.2) = 7.0143.$$

The second derivative is $f''(x) = -\frac{1}{4x^{3/2}}$, which is **negative**. Since the function is concave down, the estimate of 7.0143 is an **overestimate**.

3. Use the linear approximation to estimate $f(1.1)$ where $f(x) = x^3 - 2x + 5$. Approximate near $x = 1$.
Without computing the exact value of $f(1.1)$, is your estimate an overestimate or underestimate? Explain your answer.

The function is $f(x) = x^3 - 2x + 5$, and its first derivative is $f'(x) = 3x^2 - 2$. Near $x = 1$, we have:

$$f(1) = 1^3 - 2(1) + 5 = 4, \quad f'(1) = 3(1)^2 - 2 = 1.$$

Using the linear approximation:

$$f(1.1) \approx 4 + 1(1.1 - 1) = 4 + 0.1 = 4.1.$$

The second derivative is $f''(x) = 6x$. Since $f''(1) = 6$ is **positive**, the function is concave up, indicating that the linear approximation is an **underestimate**.

4. Approximate the value of $\ln(1.05)$ using a linear approximation near $x = 1$.
Without computing the exact value of $\ln(1.05)$, is your estimate an overestimate or underestimate? Explain your answer.

The function is $f(x) = \ln(x)$, and its first derivative is $f'(x) = \frac{1}{x}$. Near $x = 1$, we have:

$$f(1) = \ln(1) = 0, \quad f'(1) = \frac{1}{1} = 1.$$

Using the linear approximation:

$$\ln(1.05) \approx 0 + 1(1.05 - 1) = 0 + 0.05 = 0.05.$$

The second derivative is $f''(x) = -\frac{1}{x^2}$, which is **negative**. Since the function is concave down, the estimate of 0.05 is an **overestimate**.

5. Use a linear approximation to estimate $\frac{1}{2.1}$ by approximating the function $f(x) = \frac{1}{x}$ near $x = 2$.
Without computing the exact value of $\frac{1}{2.1}$, is your estimate an overestimate or underestimate? Explain your answer.

The function is $f(x) = \frac{1}{x}$, and its first derivative is $f'(x) = -\frac{1}{x^2}$. Near $x = 2$, we have:

$$f(2) = \frac{1}{2}, \quad f'(2) = -\frac{1}{2^2} = -\frac{1}{4}.$$

Using the linear approximation:

$$\frac{1}{2.1} \approx \frac{1}{2} + \left(-\frac{1}{4}\right)(2.1 - 2) = \frac{1}{2} - \frac{1}{4}(0.1) = 0.495.$$

The second derivative is $f''(x) = \frac{2}{x^3}$, which is **positive** at $x = 2$. Since the function is concave up, the estimate of 0.495 is an **underestimate**.

6. Use the linear approximation to estimate $f(0.9)$ where $f(x) = x^3+3x-2$. Approximate near $x = 1$.
Without computing the exact value of $f(0.9)$, is your estimate an overestimate or underestimate? Explain your answer.

The function is $f(x) = x^3 + 3x - 2$, and its first derivative is $f'(x) = 3x^2 + 3$. Near $x = 1$, we have:

$$f(1) = 1^3 + 3(1) - 2 = 2, \quad f'(1) = 3(1)^2 + 3 = 6.$$

Using the linear approximation:

$$f(0.9) \approx 2 + 6(0.9 - 1) = 2 - 6(0.1) = 1.4.$$

The second derivative is $f''(x) = 6x + 3$. Since $f''(1) = 9$ is **positive**, the function is concave up, indicating that the linear approximation is an **underestimate**.

7. Use the linear approximation to estimate $f(1.1)$ where $f(x) = e^x$. Approximate near $x = 1$.
Without computing the exact value of $f(1.1)$, is your estimate an overestimate or underestimate? Explain your answer.

The function is $f(x) = e^x$, and its first derivative is $f'(x) = e^x$. Near $x = 1$, we have:

$$f(1) = e^1 = e, \quad f'(1) = e^1 = e.$$

Using the linear approximation:

$$e^{1.1} \approx e + e(1.1 - 1) = e + 0.1e = 1.1e.$$

The second derivative is $f''(x) = e^x$, which is **positive**. Since the function is concave up, the estimate of $1.1e$ is an **underestimate**.

L'Hopital's Rule

Compute the following limits. Check at each step if L'Hôpital is applicable. Show all your work.

1. $\lim_{x\to 0} \dfrac{e^x - 1 - x}{x^2}$

   Check: numerator $\to 0$, denominator $\to 0$, so $0/0$ — L'Hôpital applicable.

   1st derivative:

   $$\frac{e^x - 1}{2x}.$$

   Form: $0/0$ — apply L'Hôpital again.

   2nd derivative:

   $$\frac{e^x}{2} \xrightarrow{x\to 0} \frac{1}{2}.$$

   Answer: $\boxed{\frac{1}{2}}$.

2. $\lim_{x\to 0} \dfrac{e^{2x} - 1 - 2x}{x}$

   Check: numerator $\to 0$, denominator $\to 0 \Rightarrow 0/0$ — L'Hôpital applicable.

   Differentiate numerator and denominator:

   $$\frac{2e^{2x} - 2}{1} \xrightarrow{x\to 0} 2 - 2 = 0.$$

   Answer: $\boxed{0}$.

3. $\lim_{x\to\infty} \dfrac{x}{e^x}$

   Check: ∞/∞ — L'Hôpital applicable.

   Differentiate:

   $$\frac{1}{e^x} \xrightarrow{x\to\infty} 0.$$

   Answer: $\boxed{0}$.

4. $\lim_{x\to 0^+} x \ln x$

   Rewrite as a quotient: $x \ln x = \dfrac{\ln x}{1/x}$. Check: numerator $\to -\infty$, denominator $\to +\infty \Rightarrow -\infty/\infty$ — L'Hôpital applicable.

   Differentiate numerator and denominator:

   $$\frac{1/x}{-1/x^2} = -x \xrightarrow{x\to 0^+} 0.$$

   Therefore $x \ln x \to 0$.

   Answer: $\boxed{0}$.

5. $\lim_{x\to 0} \dfrac{1-e^{-x}}{x}$

Check: numerator $\to 0$, denominator $\to 0 \Rightarrow 0/0$ — L'Hôpital applicable.

Differentiate:

$$\frac{e^{-x}}{1} \xrightarrow{x\to 0} 1.$$

Answer: $\boxed{1}$.

6. $\lim_{x\to 0} \dfrac{\ln(1+x)-x}{x^2}$

Check: numerator $\to 0$, denominator $\to 0 \Rightarrow 0/0$ — L'Hôpital applicable.

Differentiate numerator and denominator:

$$\frac{\frac{1}{1+x}-1}{2x} = \frac{\frac{-x}{1+x}}{2x} = \frac{-1}{2(1+x)}.$$

Now substitute $x \to 0$:

$$\frac{-1}{2(1+0)} = -\frac{1}{2}.$$

Answer: $\boxed{-\frac{1}{2}}$.

7. $\lim_{x\to 0} \dfrac{\ln(1+x)-x+\frac{x^2}{2}}{x^3}$

Check: numerator $\to 0$, denominator $\to 0 \Rightarrow 0/0$ — L'Hôpital applicable.

1st derivative:

$$\frac{\frac{1}{1+x}-1+x}{3x^2} \quad \text{(still } 0/0\text{)}.$$

2nd derivative:

$$\frac{-\frac{1}{(1+x)^2}+1}{6x} \quad \text{(still } 0/0\text{)}.$$

3rd derivative:

$$\frac{\frac{2}{(1+x)^3}}{6} = \frac{1}{3(1+x)^3} \xrightarrow{x\to 0} \frac{1}{3}.$$

Answer: $\boxed{\frac{1}{3}}$.

8. $\lim_{x\to 0} \dfrac{e^x-1}{x}$

Check: $0/0$ — L'Hôpital applicable.

Differentiate:

$$\frac{e^x}{1} \xrightarrow{x\to 0} 1.$$

Answer: $\boxed{1}$.

9. $\lim_{x\to \infty} \dfrac{\ln x}{x}$

Check: ∞/∞ — L'Hôpital applicable.

Differentiate:

$$\frac{1/x}{1} = \frac{1}{x} \xrightarrow{x\to\infty} 0.$$

Answer: $\boxed{0}$.

10. $\lim_{x\to 0} \dfrac{\sqrt{1+x} - 1 - \frac{x}{2}}{x^2}$

Check: numerator $\to 0$, denominator $\to 0 \Rightarrow 0/0$ — L'Hôpital applicable.

1st derivative (numerator'):

$$\frac{1}{2\sqrt{1+x}} - \frac{1}{2}.$$

So after one L'Hôpital we have

$$\frac{\frac{1}{2\sqrt{1+x}} - \frac{1}{2}}{2x}.$$

This is $0/0$ at $x = 0$ — apply L'Hôpital again.

2nd derivative (numerator"):

$$-\frac{1}{4(1+x)^{3/2}} - 0 = -\frac{1}{4(1+x)^{3/2}},$$

denominator" is 2. Hence

$$\lim_{x\to 0} \frac{-\frac{1}{4(1+x)^{3/2}}}{2} = -\frac{1}{8}.$$

Answer: $\boxed{-\frac{1}{8}}$.

11. $\lim_{x\to 0} \dfrac{1 - \sqrt{1-x}}{x}$

Check: numerator $\to 0$, denominator $\to 0 \Rightarrow 0/0$ — L'Hôpital applicable.

Differentiate numerator and denominator:

$$\frac{\frac{1}{2}(1-x)^{-1/2}}{1} = \frac{1}{2\sqrt{1-x}} \xrightarrow{x\to 0} \frac{1}{2}.$$

Answer: $\boxed{\frac{1}{2}}$.

12. $\lim_{x\to 0} \dfrac{e^x - 1 - x - \frac{x^2}{2}}{x^3}$

Check: numerator $\to 0$, denominator $\to 0 \Rightarrow 0/0$ — L'Hôpital applicable.

1st derivative:

$$\frac{e^x - 1 - x}{3x^2} \quad (0/0).$$

2nd derivative:

$$\frac{e^x - 1}{6x} \quad (0/0).$$

3rd derivative:

$$\frac{e^x}{6} \xrightarrow{x\to 0} \frac{1}{6}.$$

Answer: $\boxed{\frac{1}{6}}$.

13. $\lim_{x\to\infty} \dfrac{e^x}{x^2}$

Check: ∞/∞ — L'Hôpital applicable.

1st derivative:

$$\frac{e^x}{2x} \text{ (still } \infty/\infty).$$

2nd derivative:

$$\frac{e^x}{2} \xrightarrow{x\to\infty} \infty.$$

Answer: $\boxed{+\infty}$.

14. $\lim_{x\to 0} \dfrac{e^x + 1}{x}$

Check: numerator $\to 2$, denominator $\to 0$ — not $0/0$ or ∞/∞, so L'Hôpital does not apply.

Right-hand limit $+\infty$, left-hand limit $-\infty$; two-sided limit does not exist.

Answer: two-sided limit does not exist; $\lim_{x\to 0^+} = +\infty$, $\lim_{x\to 0^-} = -\infty$.

15. $\lim_{x\to 0} \dfrac{e^x - 1 - x - \frac{x^2}{2} - \frac{x^3}{6}}{x^4}$

Check: numerator $\to 0$, denominator $\to 0 \Rightarrow 0/0$ — L'Hôpital applicable.

Apply L'Hôpital repeatedly:

1st derivative:

$$\frac{e^x - 1 - \frac{x^2}{2} - \frac{x^3}{6}}{4x^3} \quad (0/0).$$

2nd derivative:

$$\frac{e^x - x - \frac{x^2}{2}}{12x^2} \quad (0/0).$$

3rd derivative:

$$\frac{e^x - 1 - x}{24x} \quad (0/0).$$

4th derivative:

$$\frac{e^x}{24} \xrightarrow{x\to 0} \frac{1}{24}.$$

Answer: $\boxed{\frac{1}{24}}$.

Related Rates

1. **Inflating a Spherical Balloon**
   Air is pumped into a spherical balloon so that its surface area increases at 24 cm^2/s.
   (a) How fast is the radius increasing when the radius is 3 cm?
   (b) How fast is its volume changing at that same instant?

   (a) $A = 4\pi r^2$, so
   $$\frac{dA}{dt} = 8\pi r \frac{dr}{dt}.$$
   Thus
   $$\frac{dr}{dt} = \frac{\frac{dA}{dt}}{8\pi r} = \frac{24}{8\pi \cdot 3} = \frac{1}{\pi} \text{ cm/s}.$$

   (b) $V = \frac{4}{3}\pi r^3$, so
   $$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}.$$
   At $r = 3$,
   $$\frac{dV}{dt} = 4\pi(3)^2 \left(\frac{1}{\pi}\right) = 36 \text{ cm}^3/\text{s}.$$

2. **Water Filling a Conical Tank**
   Water fills a conical tank.

   (a) Explain why, if the radius is 2 ft when the water depth is 6 ft, it is always true that $r = \frac{1}{3}h$.

   Similar triangles: since the internal angles are the same (at the point of the cone), and the radius is always perpendicular to the depth, the triangles are similar, and so the radius and height always have the same ratio.

   (b) The volume increases at 2 ft^3/min. How fast is the height rising when $h = 6$ ft?

   $V = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi \left(\frac{h}{3}\right)^2 h = \frac{1}{27}\pi h^3$. Differentiate:
   $$\frac{dV}{dt} = \frac{1}{9}\pi h^2 \frac{dh}{dt}.$$
   Thus
   $$\frac{dh}{dt} = \frac{\frac{dV}{dt}}{\frac{1}{9}\pi h^2} = \frac{2}{\frac{1}{9}\pi(6)^2} = \frac{1}{2\pi} \text{ ft/min}.$$

3. **Expanding Circular Puddle**
   A circular puddle's area is increasing at 10 m^2/hr. When the radius is 5 m, how fast is the radius increasing?

   $A = \pi r^2$, so
   $$\frac{dA}{dt} = 2\pi r \frac{dr}{dt}.$$
   Hence
   $$\frac{dr}{dt} = \frac{\frac{dA}{dt}}{2\pi r} = \frac{10}{2\pi \cdot 5} = \frac{1}{\pi} \text{ m/hr}.$$

4. **Rectangle With One Side Fixed**
A poster has height 12 in and width increasing at 0.5 in/s. How fast is its area increasing when the width is 8 in?

$A = 12w$, so

$$\frac{dA}{dt} = 12\frac{dw}{dt} = 12(0.5) = 6 \text{ in}^2/\text{s}.$$

5. **Growing Cylinder**
A Cylinder has radius 2 m at a certain instant. Its radius is growing at 0.03 m/min. The water depth is always 3 m. How fast is its volume changing?

$V = \pi r^2 h$, so

$$\frac{dV}{dt} = 2\pi rh\frac{dr}{dt}.$$

Thus

$$\frac{dV}{dt} = 2\pi(2)(3)(0.03) = 0.36\pi \text{ m}^3/\text{min}.$$

6. **Two Cars**
Car A moves east at 60 mph and Car B north at 40 mph. At one moment, A is 3 miles east and B is 4 miles north of the intersection. How fast is the distance between them changing?

$s^2 = x^2 + y^2$. Differentiate:

$$2s\frac{ds}{dt} = 2x\frac{dx}{dt} + 2y\frac{dy}{dt}.$$

So

$$\frac{ds}{dt} = \frac{x\frac{dx}{dt} + y\frac{dy}{dt}}{s}.$$

Since $s = 5$ (Pythagoras),

$$\frac{ds}{dt} = \frac{3 \cdot 60 + 4 \cdot 40}{5} = 68 \text{ mph}.$$

7. **Inflating Cube**
A cube's surface area increases at 30 cm^2/s. How fast is the edge increasing, and how fast is the volume increasing, when the edge is 5 cm?

$A = 6s^2$, so

$$\frac{dA}{dt} = 12s\frac{ds}{dt}.$$

Thus

$$\frac{ds}{dt} = \frac{\frac{dA}{dt}}{12s} = \frac{30}{12 \cdot 5} = \frac{1}{2}.$$

Volume $V = s^3$, so

$$\frac{dV}{dt} = 3s^2\frac{ds}{dt} = 3(5)^2\left(\frac{1}{2}\right) = 37.5 \text{ cm}^3/\text{s}.$$

8. **Rectangle with Fixed Diagonal**
A rectangle has diagonal 13 cm. Its width increases at 0.2 cm/s. How fast is the area changing when the width is 5 cm?

(Hint: relate the width and height to the length of the diagonal)

$w^2 + h^2 = 13^2$. Differentiate:

$$2w\,\frac{dw}{dt} + 2h\,\frac{dh}{dt} = 0,$$

so

$$\frac{dh}{dt} = -\frac{w}{h}\,\frac{dw}{dt}.$$

At $w = 5$, $h = 12$. Thus

$$\frac{dh}{dt} = -\frac{5}{12}(0.2) = -\frac{1}{12}.$$

Area $A = wh$, so

$$\frac{dA}{dt} = w\,\frac{dh}{dt} + h\,\frac{dw}{dt}.$$

Hence

$$\frac{dA}{dt} = 5\left(-\frac{1}{12}\right) + 12(0.2) = -\frac{5}{12} + 2.4 = \frac{119}{60} \approx 1.98333 \text{ cm}^2/\text{s}.$$